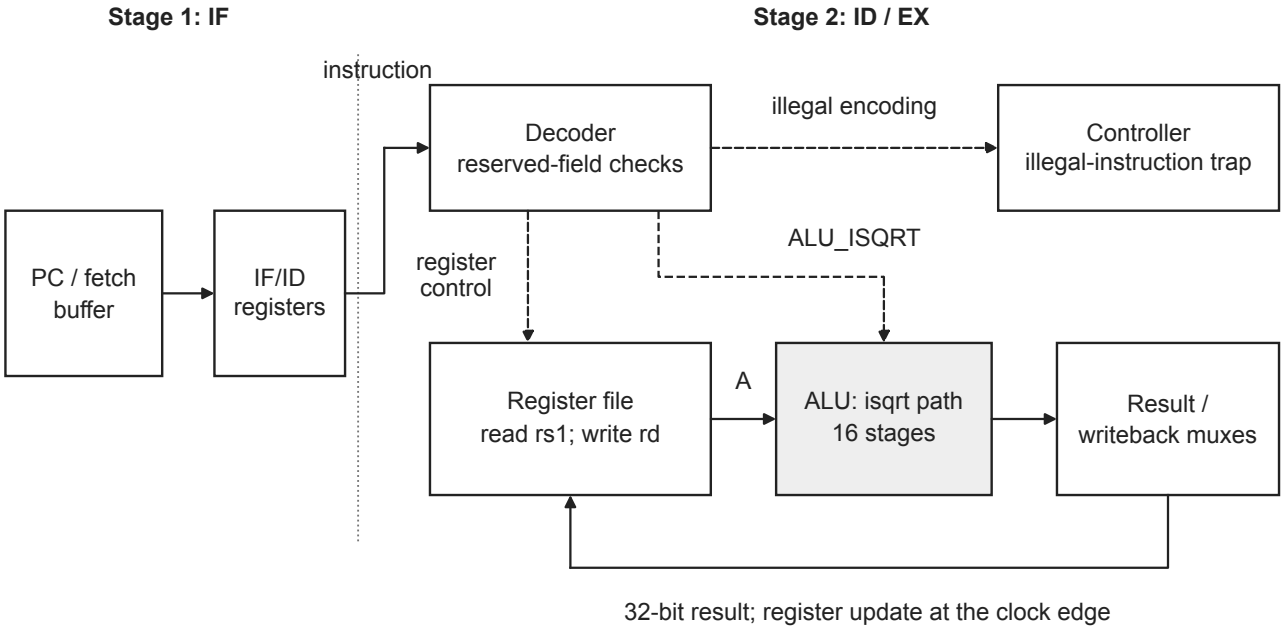
Stage 1: IF
Stage 2: ID / EX
instruction
PC / fetch buffer
IF/ID registers
Decoder reserved-field checks
illegal encoding
Controller illegal-instruction trap
register control
ALU_ISQRT
Register file read rs1; write rd
A
ALU: isqrt path 16 stages
Result / writeback muxes
32-bit result; register update at the clock edge